

TED (15)6121

Reg.No.-----

(Revision 2015)

Signature-----

SIXTH SEMESTER DIPLOMA IN WOOD AND PAPER TECHNOLOGY

ADHESIVE TECHNOLOGY

(Maximum marks:100)

Time:3Hrs

PART-

Marks

(Maximum marks:10)

I. Answer all questions in one or two sentences. Each question carries 2 marks.

1. Define the term rheology
2. Define water tolerance of resin
3. State amino resin
4. State polymerization
5. List any two thermosetting resin (5X2=10)

PART.B

(Maximum Marks:30)

II. Answer any five of the following questions. Each question carry 6 Marks

1. Describe Cohesion and Adhesion
2. Describe the phenomenon of hardening
3. Describe silicate of soda as a glue
4. Describe the chemistry of UF resin
5. Describe phenolic resin
6. Describe Isocyanate resin
7. Explain cellulose glues (6x5=30)

PART.C

(Maximum Marks:60)

(Answer one full question from each unit. Each full question Each full question carry 15 Marks)

Module. I

III. (a) Describe the following properties of glue

(a) Water tolerance (b) Viscosity (c) Solid content 9

(b) Describe the intermolecular forces between atoms and molecules 6

OR

IV (a) Describe roller spreading of glue 9

(b) Describe the nature of wood species in gluing 6

Module. I

V Write short notes on

a) Natural rubber (b) Starch glue (c) Cellulose ester (6+5+4=15)

OR

VI Describe the any 5 Additives used in Glue mixing 15

Module.III

VII (a) Describe the method of manufacture of UF resin 9

(b) Describe the method of manufacturing of UMF resin 6

OR

VIII (a) Describe the industrial application of MF resin 9

(b) Draw a neat sketch of resin kettle and label the parts 6

Module.IV

X.	(a) Describe the method of manufacture of PF resin	8
	(b) Describe the raw materials used in poly urethane adhesives	7
	OR	
X	(a) Describe the advantage and disadvantage of PF resins	8
	(b) Describe the method of manufacture of Epoxy resin	7